
Bayesian Networks: Sampling

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: [Prof. Wei](#)'s CS4486 Course
and [Prof. Boddeti](#)'s AI Course



TODAY

Prior Sampling

Rejection Sampling

Likelihood Weighting

Gibbs Sampling



RECAP: BAYES NET REPRESENTATION

A directed, acyclic graph, one node per random variable

A conditional probability table (CPT) for each node

- A collection of distributions over $\mathbf{X}$, one for each combination of parents values

$$P(\mathbf{X} | a_1, \dots, a_n)$$

Bayes nets implicitly encode joint distributions

- As a product of local conditional distributions
- To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(x_i))$$



RECAP: VARIABLE ELIMINATION

Interleave joining and marginalizing

d^k entries computed for a factor over k variables with domain sizes d

Ordering of elimination of hidden variables can affect size of factors generated

Worst case: running time exponential in the size of the Bayes net



APPROXIMATE INFERENCE: SAMPLING



SAMPLING

Why sample?



SAMPLING

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- Learning: get samples from a distribution you don't know
- Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



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Sampling is a lot like repeated simulation

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- Predicting the weather, basketball games, . . .

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Basic idea

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Basic idea

- Draw N samples from a sampling distribution $\mathcal{S}$
- Compute an approximate posterior probability
- Show this converges to the true probability P

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SAMPLING

Sampling from given distribution



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- Step 1: Get sample u from uniform distribution over $[0, 1)$
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Example:

C	$P(C)$
<i>red</i>	0.6
<i>green</i>	0.1
<i>blue</i>	0.3

$$0 \leq u < 0.6, \rightarrow C = \textit{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \textit{green}$$

$$0.7 \leq u < 1.0, \rightarrow C = \textit{blue}$$



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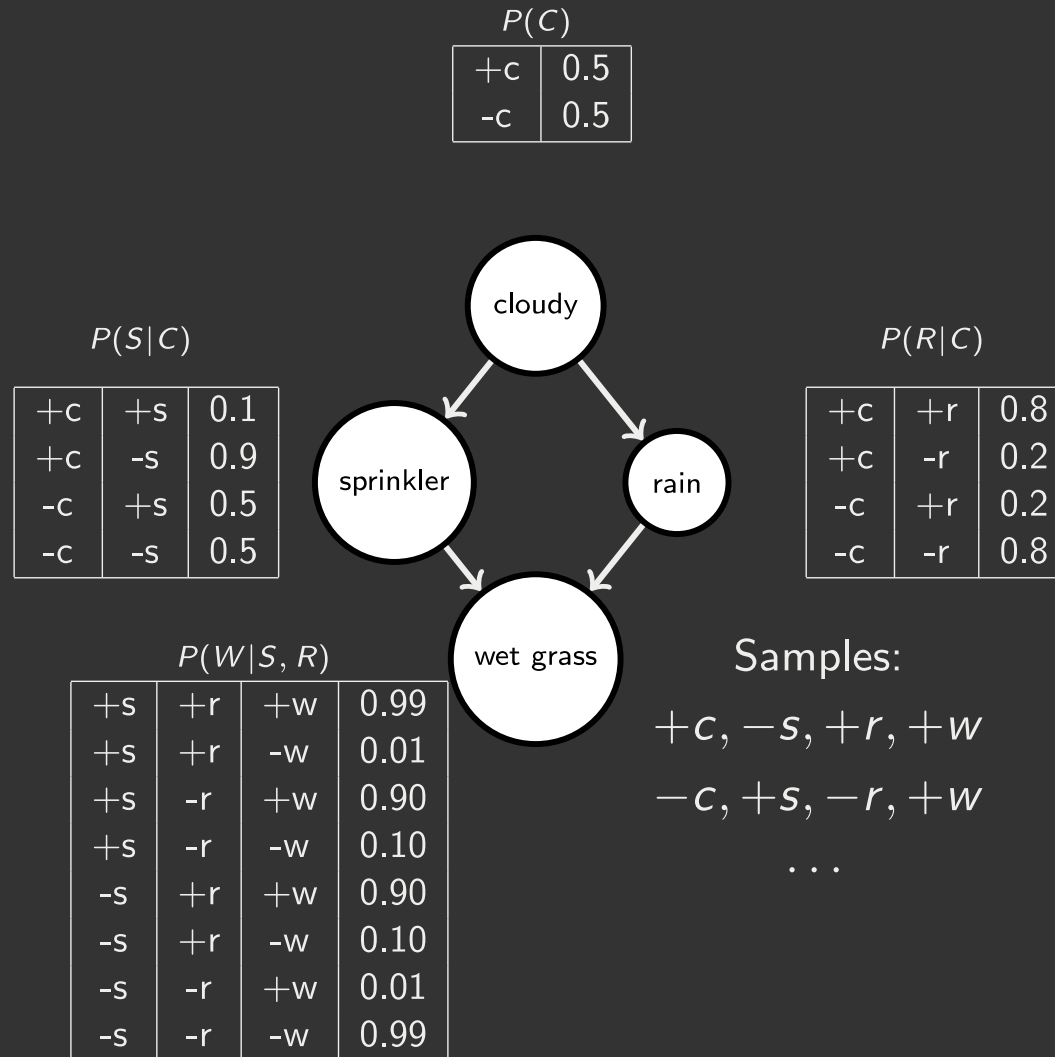
- If *random()* returns $u = 0.83$, then our sample is $C = \textit{blue}$.



PRIOR SAMPLING



PRIOR SAMPLING



PRIOR SAMPLING

Prior Sampling:

```
Function prior-sampling(bayes net)  
  for  $i = 1, 2, \dots, n$  do  
    | Sample  $x_i$  from  $P(X_i | Parents(X_i))$  ;  
    | return  $(x_1, x_2, \dots, x_n)$ ;  
  end
```



PRIOR SAMPLING

This process generates samples with probability:

$$S_{PS}(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1, \dots, x_n)$$

i.e., the BN's joint probability.



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Then,

$$\begin{aligned} \lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} \frac{N_{PS}(x_1, \dots, x_n)}{N} \\ &= S_{PS}(x_1, \dots, x_n) \\ &= P(x_1, \dots, x_n) \end{aligned}$$



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i.e., the sampling procedure is **consistent**



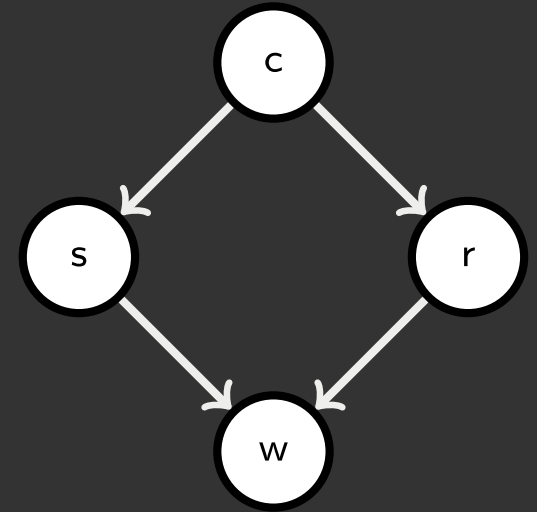
EXAMPLE: PRIOR SAMPLING

We will get a bunch of samples from the BN:

- $+c, -s, +r, +w$
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- $-c, +s, +r, -w$
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If we want to know $P(W)$

- We have counts $\langle +w : 4, -w : 1 \rangle$
- Normalize to get $P(W) = \langle +w : 0.8, -w : 0.2 \rangle$
- This will get closer to the true distribution with more samples
- Can estimate anything else, too



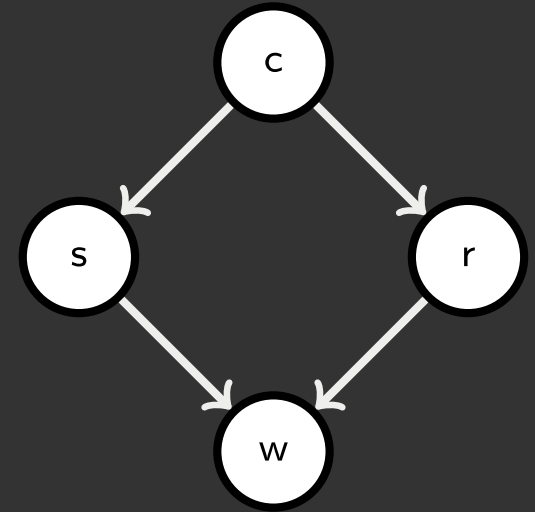
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- What about $P(C | +w)$? $P(C | +r, +w)$? $P(C | -r, -w)$?



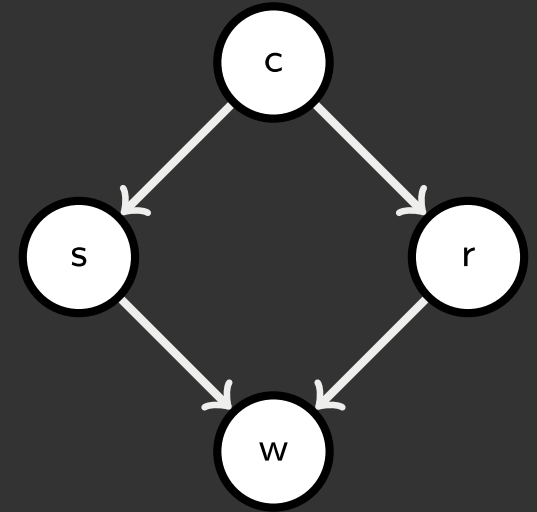
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- Can estimate anything else, too
- What about $P(C | +w)$? $P(C | +r, +w)$? $P(C | -r, -w)$?
- Fast: can use fewer samples if less time (what is the drawback?)



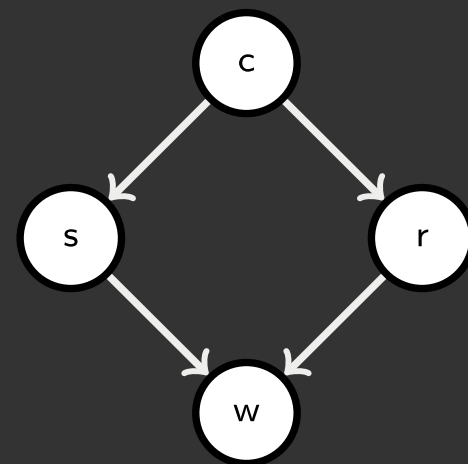
REJECTION SAMPLING



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Let us say we want $P(C)$

- No point keeping all samples around
- Just tally counts of C as we go



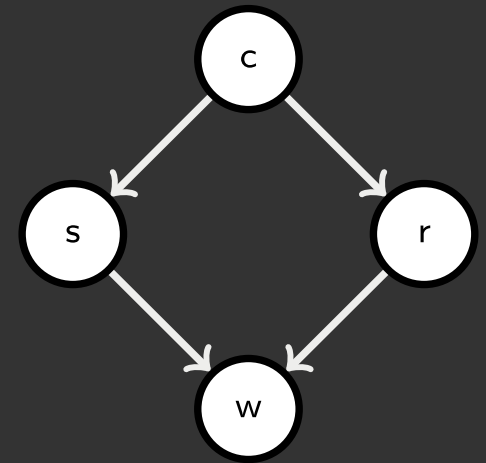
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Let us say we want $P(C | +s)$



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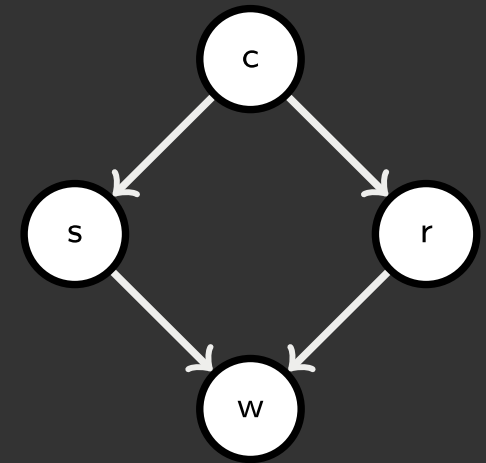
REJECTION SAMPLING

Let us say we want $P(C)$

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Let us say we want $P(C | +s)$

- Same thing: tally C outcomes, but ignore (reject) samples which do not have $S = +s$
- This is called [rejection sampling](#)
- It is also consistent for conditional probabilities (i.e., correct in the limit)



$+c, -s, +r, +w$
 $+c, +s, +r, +w$
 $-c, +s, +r, -w$
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REJECTION SAMPLING

Rejection Sampling:

```
Function rejection-sampling(bayes-net,  
    evidence-instantiation)  
    for  $i = 1, 2, \dots, n$  do  
        Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        if  $x_i$  not consistent with evidence then  
            | Reject: return – no sample is generated in this cycle  
        end  
        return  $(x_1, x_2, \dots, x_n)$ ;  
    end
```



LIKELIHOOD WEIGHTING



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Problem with rejection sampling:



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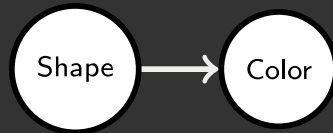
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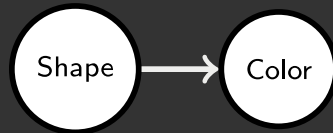
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- Consider $P(\textit{Shape}|\textit{blue})$



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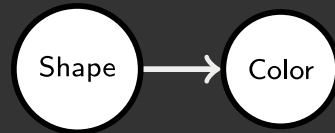
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Idea: fix evidence variables and sample the rest

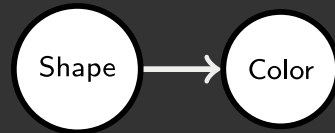


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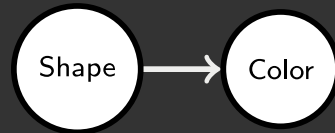
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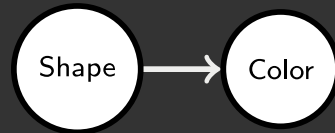
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- Problem: sample distribution not consistent
- Solution: weight by probability of evidence given parents

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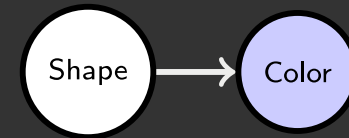
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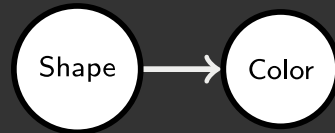
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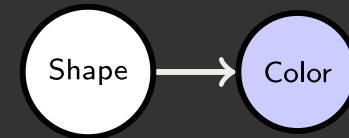
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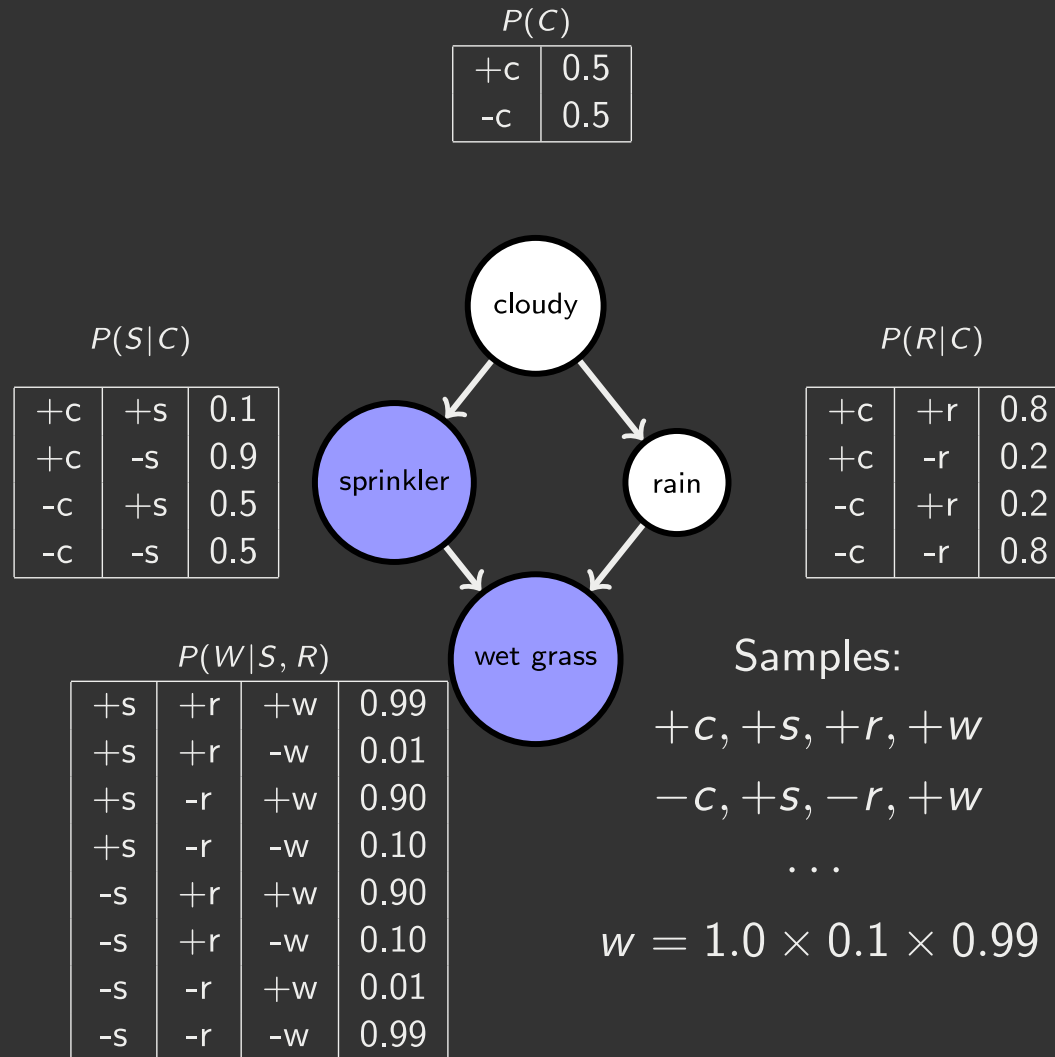
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LIKELIHOOD WEIGHTING: EXAMPLE



LIKELIHOOD WEIGHTING: ALGORITHM

Likelihood Weighting:

```
Function likelihood-weighting(bayes-net,  
    evidence-instantiation)  
     $w = 1.0$  ;  
    for  $i = 1, 2, \dots, n$  do  
        Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        if  $x_i$  is an evidence variable then  
             $X_i = \text{observation } x_i \text{ for } X_i$  ;  
            Set  $w = w * P(x_i | \text{Parents}(X_i))$  ;  
        end  
        else  
            Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        end  
    return  $(x_1, x_2, \dots, x_n), w$  ;  
end
```



LIKELIHOOD WEIGHTING

Sampling distribution if $\mathbf{z}$ sampled and $\mathbf{e}$ fixed evidence

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$



LIKELIHOOD WEIGHTING

Sampling distribution if $\mathbf{z}$ sampled and $\mathbf{e}$ fixed evidence

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$

Now, samples have weights

$$w(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^m P(e_i | \text{Parents}(E_i))$$



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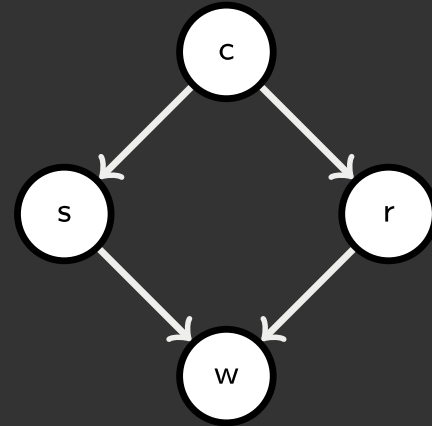
Together, weighted sampling distribution is consistent

$$\begin{aligned} S_{WS}(z, e) \cdot w(z, e) &= \prod_{i=1}^l P(z_i | Parents(z_i)) \prod_{i=1}^l P(e_i | Parents(e_i)) \\ &= P(\mathbf{z}, \mathbf{e}) \end{aligned}$$



LIKELIHOOD WEIGHTING

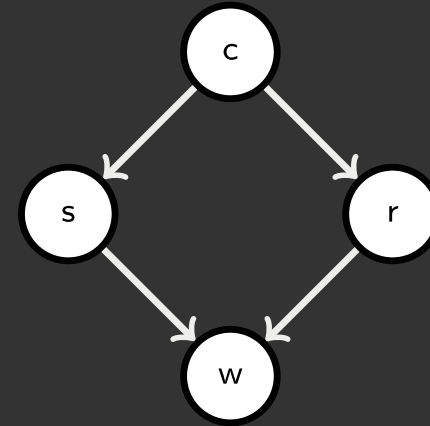
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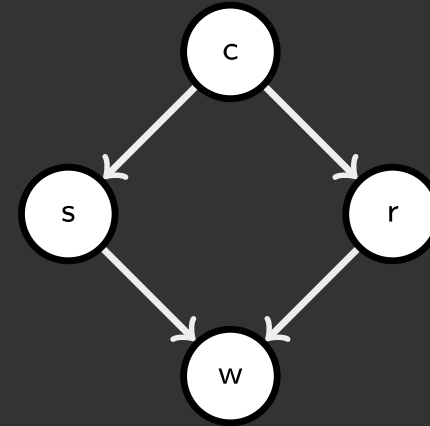
- We have taken evidence into account as we generate the sample
- E.g. here, W 's value will get picked based on the evidence values of S , R
- More of our samples will reflect the state of the world suggested by the evidence



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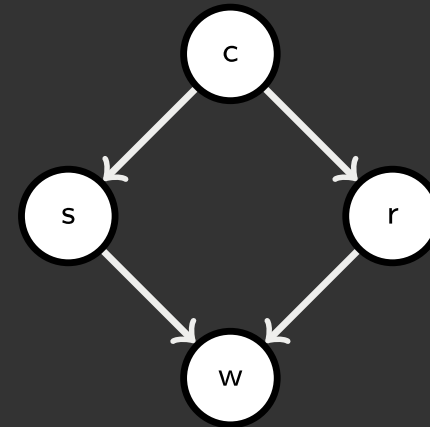


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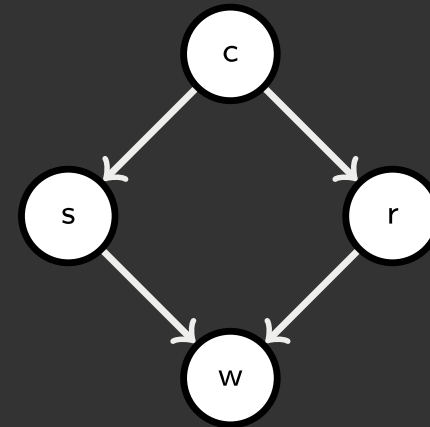
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- Evidence influences the choice of downstream variables, but not upstream ones (C is not more likely to get a value matching the evidence)

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We would like to consider evidence when we sample every variable (leads to Gibbs sampling)

GIBBS SAMPLING



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Procedure: keep track of a full instantiation $x_1, x_2, \dots, x_n$. Start with an arbitrary instantiation consistent with the evidence. Sample one variable at a time, conditioned on all the rest, but keep evidence fixed. Keep repeating this for a long time.



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Property: in the limit of repeating this infinitely many times the resulting samples come from the correct distribution (i.e. conditioned on evidence) variable.



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In contrast: likelihood weighting only conditions on upstream evidence, and hence weights obtained in likelihood weighting can sometimes be very small. Sum of weights over all samples is indicative of how many "effective" samples were obtained, so we want high weight.



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

Step 1: Fix evidence



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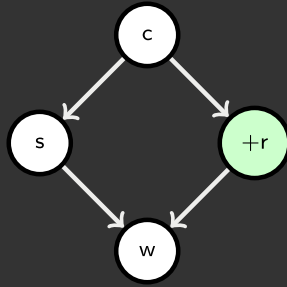
- $R = +r$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

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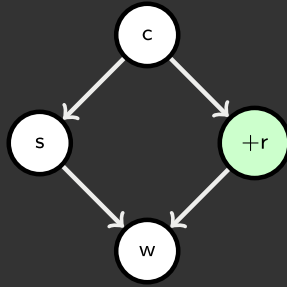
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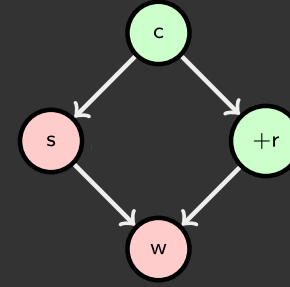
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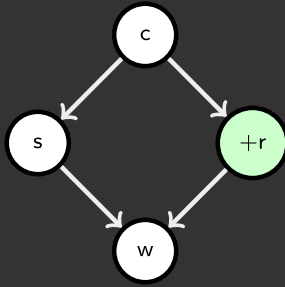
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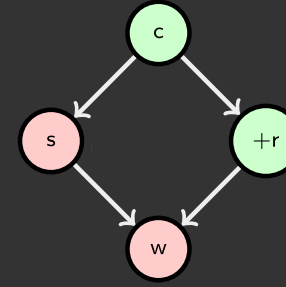
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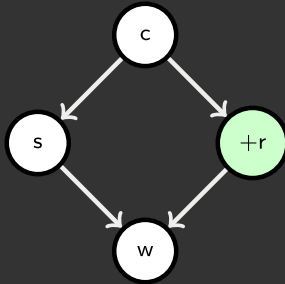


Step 3: Repeat

GIBBS SAMPLING EXAMPLE: $P(S|+R)$

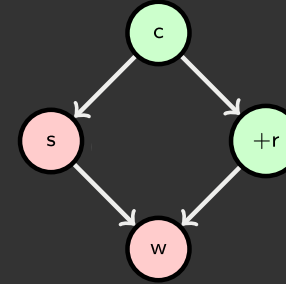
Step 1: Fix evidence

- $R = +r$



Step 2: Initialize other variables

- randomly



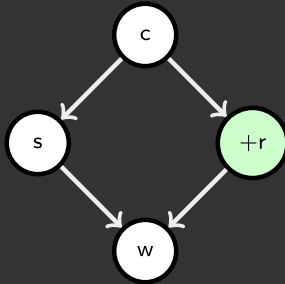
Step 3: Repeat

- Choose a non-evidence variable X

GIBBS SAMPLING EXAMPLE: $P(S|+R)$

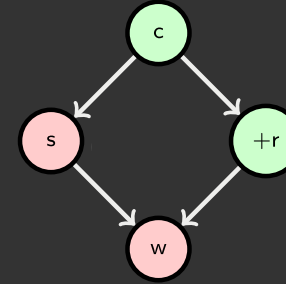
Step 1: Fix evidence

- $R = +r$



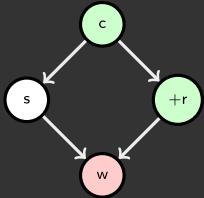
Step 2: Initialize other variables

- randomly



Step 3: Repeat

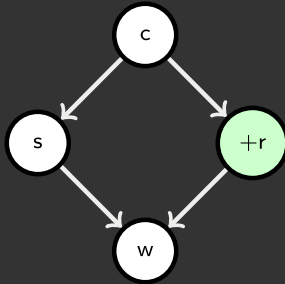
- Choose a non-evidence variable X



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

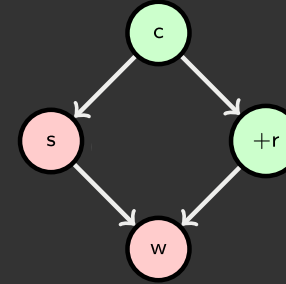
Step 1: Fix evidence

- $R = +r$



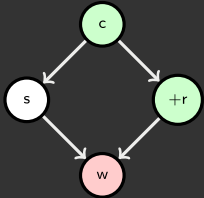
Step 2: Initialize other variables

- randomly



Step 3: Repeat

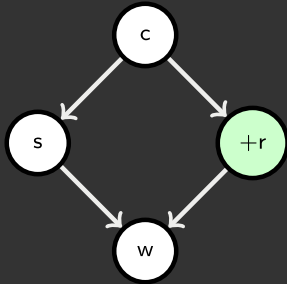
- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

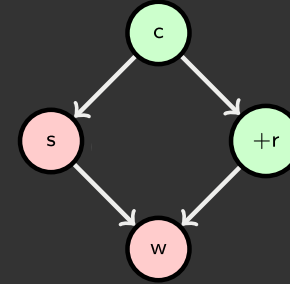
Step 1: Fix evidence

- $R = +r$



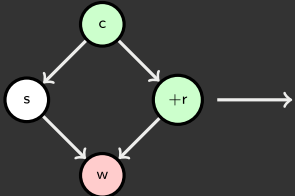
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



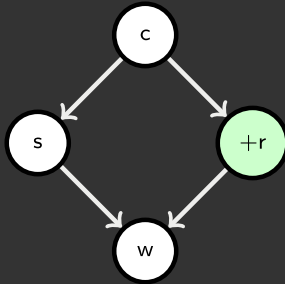
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

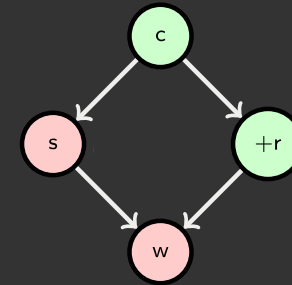
Step 1: Fix evidence

- $R = +r$



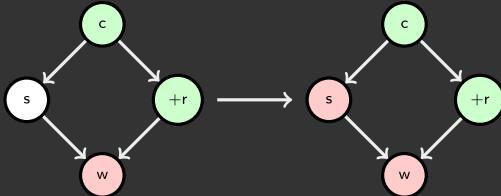
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



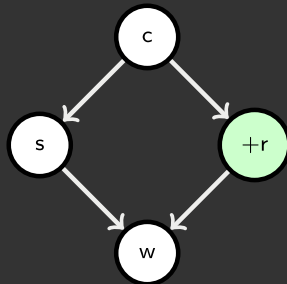
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

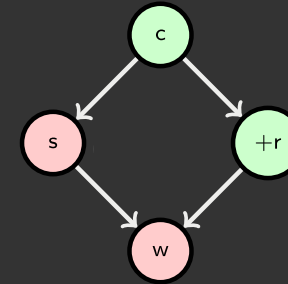
Step 1: Fix evidence

- $R = +r$



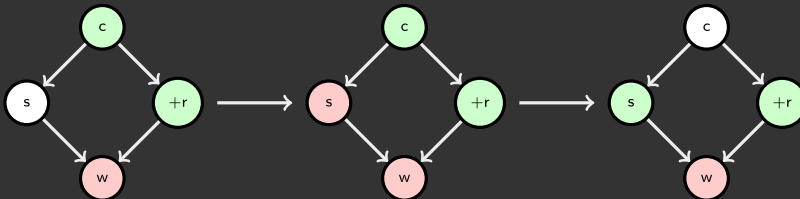
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



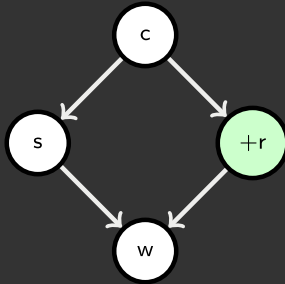
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

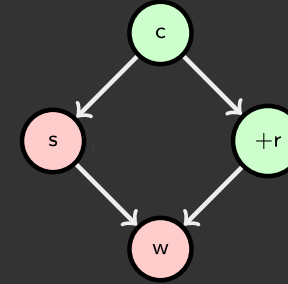
Step 1: Fix evidence

- $R = +r$



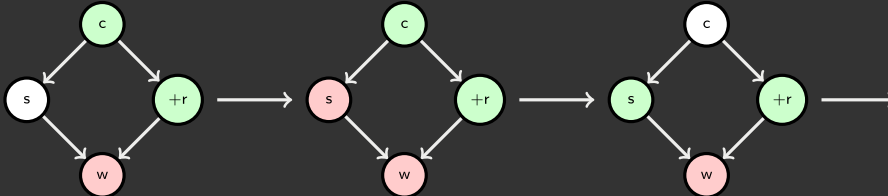
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

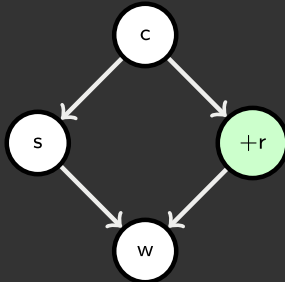
Sample from $P(C|+s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

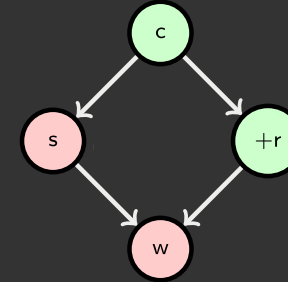
Step 1: Fix evidence

- $R = +r$



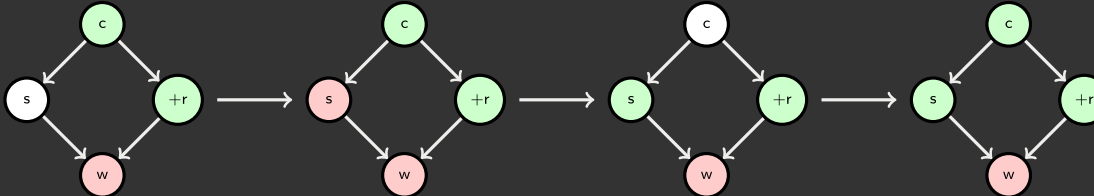
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

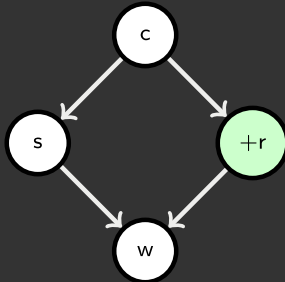
Sample from $P(C|+s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

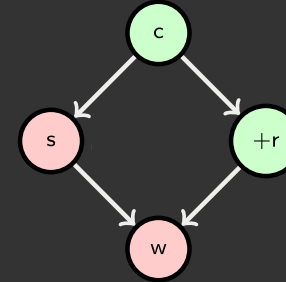
Step 1: Fix evidence

- $R = +r$



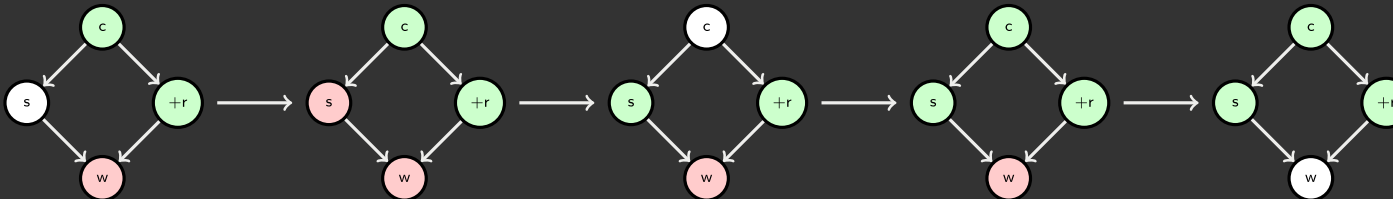
Step 2: Initialize other variables

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Sample from $P(S|+c, -w, +r)$

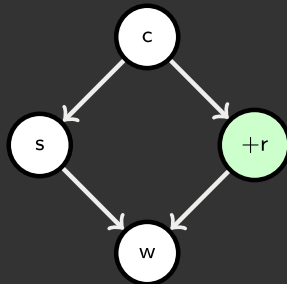
Sample from $P(C|+s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

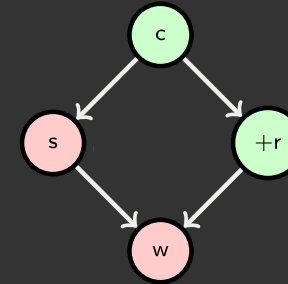
Step 1: Fix evidence

- $R = +r$



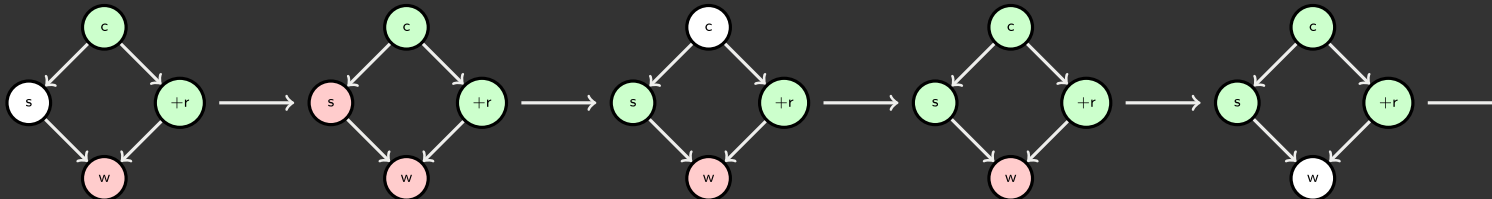
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

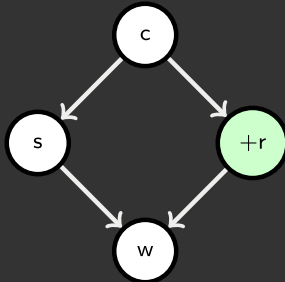
Sample from $P(C|+s, -w, +r)$

Sample from $P(W|+s, +c, +r)$

GIBBS SAMPLING EXAMPLE: $P(S|+R)$

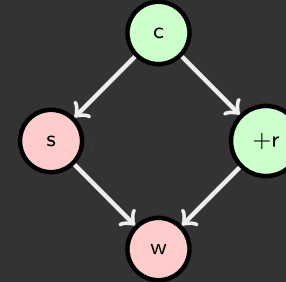
Step 1: Fix evidence

- $R = +r$



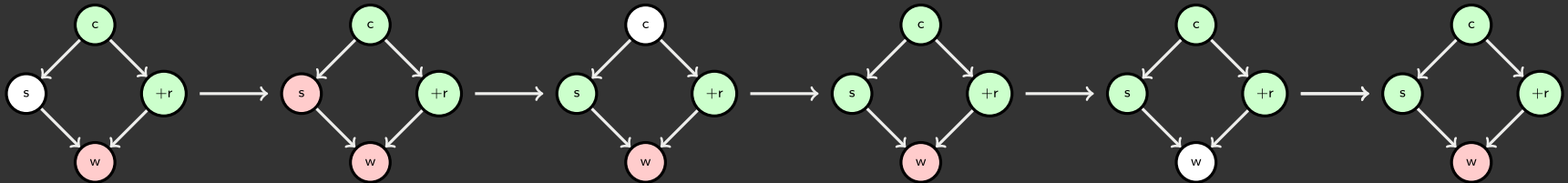
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

Sample from $P(C|+s, -w, +r)$

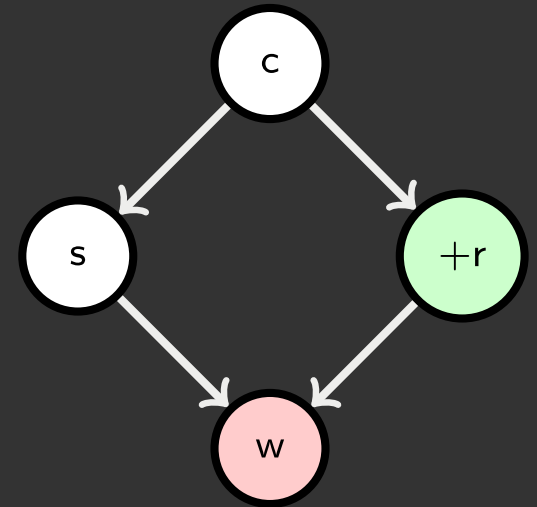
Sample from $P(W|+s, +c, +r)$



EFFICIENT RESAMPLING OF ONE VARIABLE

Sample from $P(S \mid +c, +r, -w)$

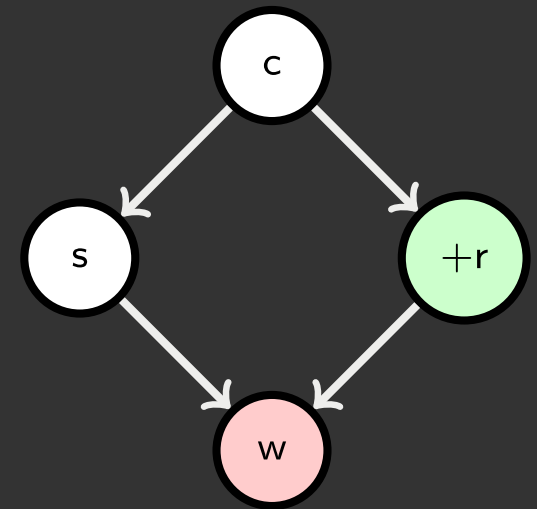
$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(S \mid +c)P(-w \mid s, +r)} \end{aligned}$$



EFFICIENT RESAMPLING OF ONE VARIABLE

Sample from $P(S \mid +c, +r, -w)$

$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(S \mid +c)P(-w \mid s, +r)} \end{aligned}$$



Many things cancel out – only CPTs with S remain

More generally: only CPTs that have resampled variable need to be considered, and joined together

SUMMARY: BAYES NET SAMPLING

Prior Sampling $P(Q)$

Rejection Sampling $P(Q|e)$

Likelihood Weighting $P(Q|e)$

Gibbs Sampling $P(Q|e)$



BEYOND GIBBS SAMPLING

Gibbs sampling produces sample from the query distribution $P(Q|e)$ in limit of re-sampling infinitely often



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Gibbs sampling is a special case of more general methods called Markov chain Monte Carlo (MCMC) methods

- Metropolis-Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis-Hastings)



BEYOND GIBBS SAMPLING

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Gibbs sampling is a special case of more general methods called Markov chain Monte Carlo (MCMC) methods

- Metropolis-Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis-Hastings)

You may read about Monte Carlo methods – they are just sampling



Q & A



XKCD

